

PROJECT PROPOSAL

PROJECT TITLE

Final Year Project Management System (FYPMS)

PROBLEM STATEMENT

At a higher learning institution, the final year project portal allows students to submit project titles and objectives electronically, but the system relies on manual checking to detect duplicate projects. This manual process is slow, time-consuming, and prone to human errors, which may result in repeated or very similar project topics being approved.

MAIN OBJECTIVE

To develop a university final year project management system that automatically performs duplicate and similarity detection of project titles and objectives.

SPECIFIC OBJECTIVES

1. To design a databases system to store project details, mentor assignment and student data.
2. To implement duplicate check algorithms that automatically detect similar project titles and objectives by integrating TF-IDF, Levenshtein and SBERT
3. To implement system mechanism that preprocess and store project titles and objectives for hybrid similarity detection.
4. To develop a feature that allows students to view other students

assigned to the same mentor.

LITERATURE REVIEW

Existing systems manage final year project activities online to improve efficiency. However, they still have gaps that the proposed system aims to solve.

Review of existing studies and systems

The National Institute of Transport (NIT) Final Year Project Portal is a web-based system that allows students to submit project titles, objectives, and descriptions online, while viewing mentor assignments, panel comments, and marks. It was developed to reduce manual submission challenges and improve project tracking using web technologies and a database system.

Soyemi & Isinkaye (2017) developed an academic project duplication detection system to automatically identify repeated or highly similar project titles. Where new project titles are compared against existing titles in a MySQL database using SQL queries. The system detects duplication by matching keywords and structure to alert supervisors about repeated projects.

Raznida Isa et al. (2025) proposed a centralized online project management platform for final year projects. This system allowed supervisors to monitor student progress, track submissions, and access all project records from a single platform. It solved the inefficiency of manual project tracking and used web technologies for data management and user interaction.

Identified gaps in existing system

From the reviewed literature, it is evident that many existing final year project portals focus mainly on project submission and record keeping. However, most systems lack:

- Automated duplicate or similarity detection for project titles and objectives
- Hybrid similarity analysis using methods like Levenshtein, TF-IDF, and SBERT
- Access to past approved projects for multiple years

These limitations reduce the effectiveness of the systems in ensuring originality, efficient tracking, and proper supervision of student projects.

SCOPE OF THE STUDY

Project will focus on design and implementation of web-based project management system and duplicate detection system for final year in National Institute of Transport (NIT) CCT department. The project will also cover a three-years period, including the current academic year.

SIGNIFICANCE OF THE PROJECT

University final year project management system will help to ensure originality by automatically detecting duplicate or similar project titles and objectives, saving time for administrators by reducing manual checks.

It also improves coordination by allowing students to view fellow students assigned to the same mentor.

Additionally, it enhances overall project management by storing past projects and tracking submissions efficiently, promoting organization and better supervision within the portal.

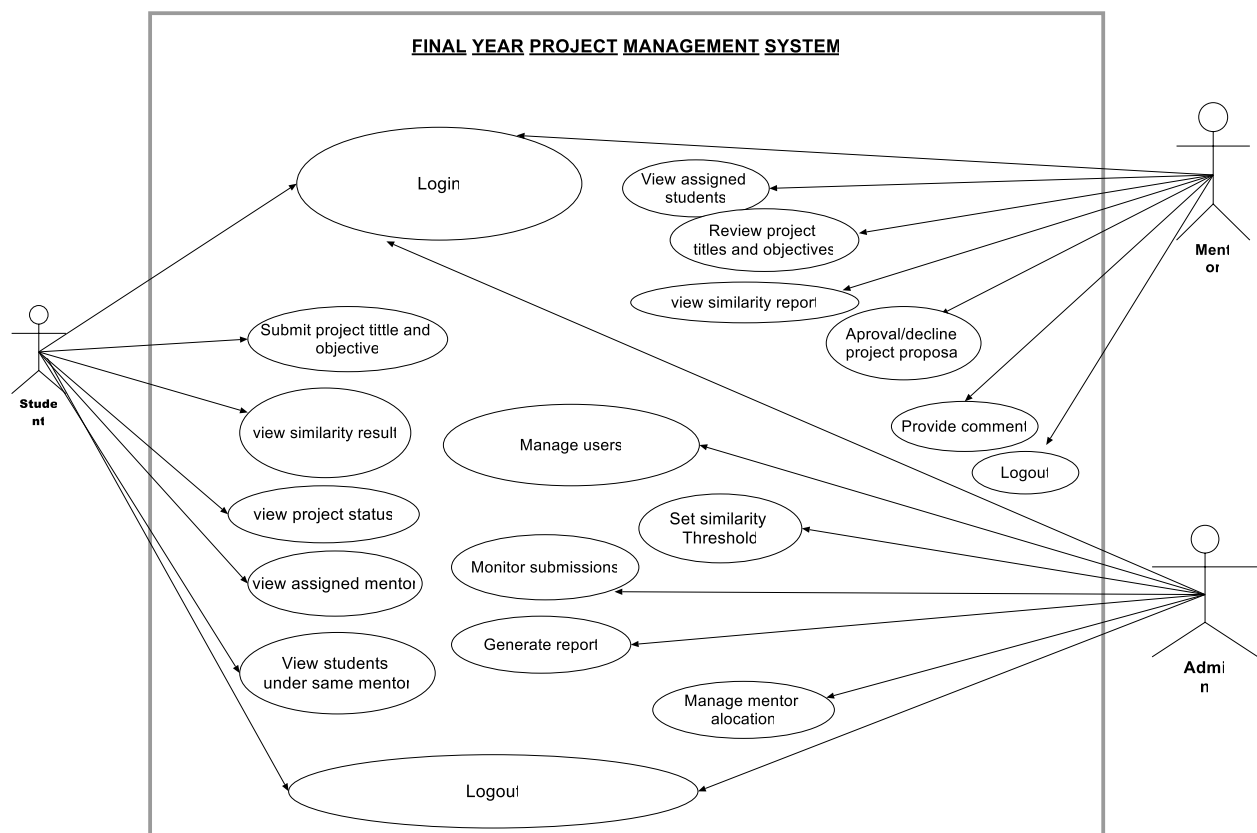
SYSTEM DESIGN

The Project Design of Final Year Project Management System

It includes a Use Case Diagram to illustrate user interactions and a Data Flow Diagram to show how data moves within the system.

These diagrams explain the overall functionality and processes of the system.

USE CASE DIAGRAM OF THE FINAL YEAR PROJECT MANAGEMENT SYSTEM



FLOWCHART DIAGRAM OF THE FINAL YEAR PROJECT MANAGEMENT SYSTEM

